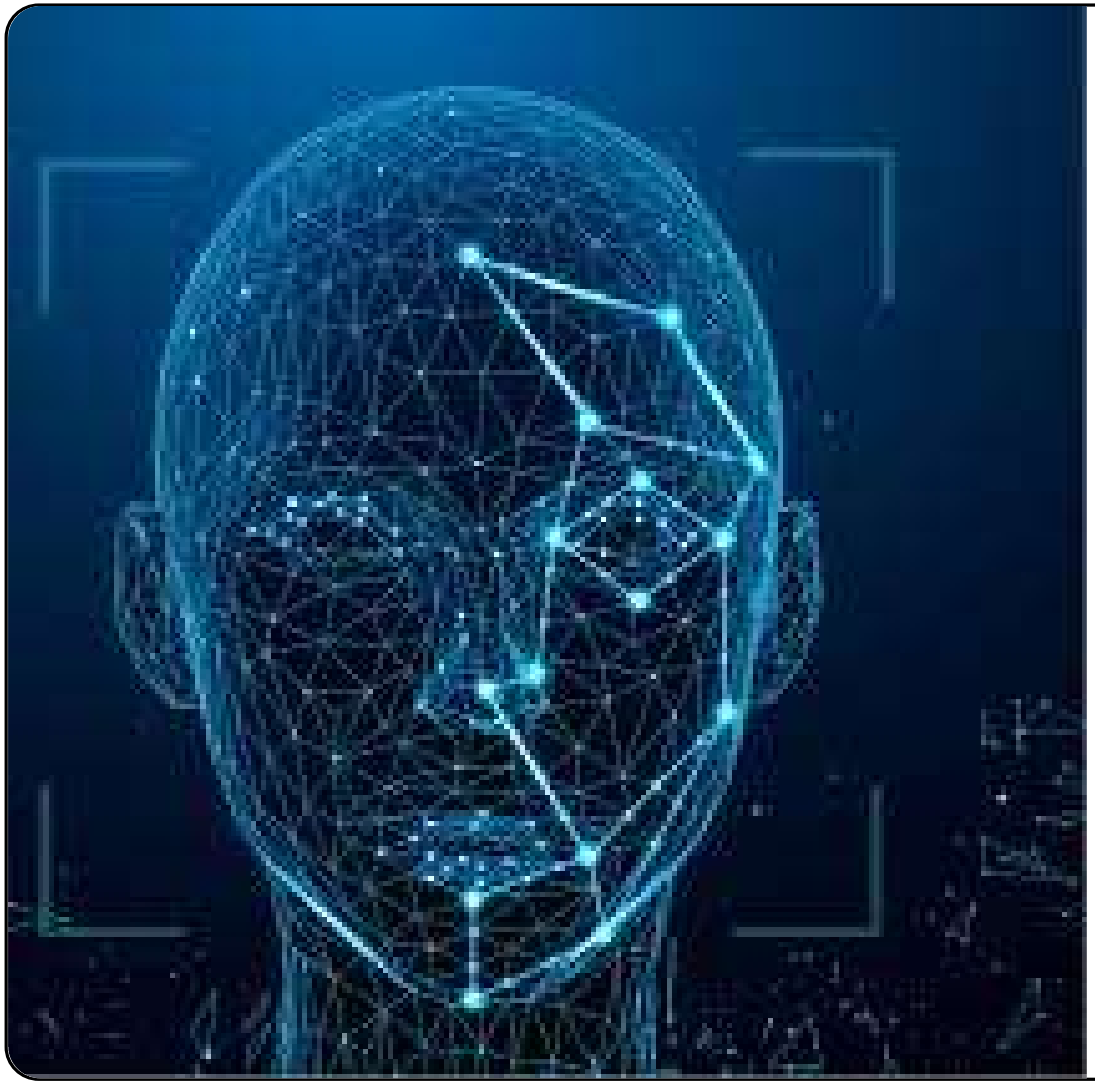

EMOTION RECOGNITION PROJECT

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CSE 2600: Introduction to Data Science and Engineering

BACKGROUND, GOALS, AND MOTIVATION

Motivation:

- Facial emotion classification has real-world applications in human-computer interaction, accessibility technology, and mental health support systems
- We wanted to explore how data science methods learned in class, especially PCA and Random Forest, can be applied to facial emotion classification
- We were interested in whether dimensionality reduction could simplify the feature space while still preserving useful emotion-related information

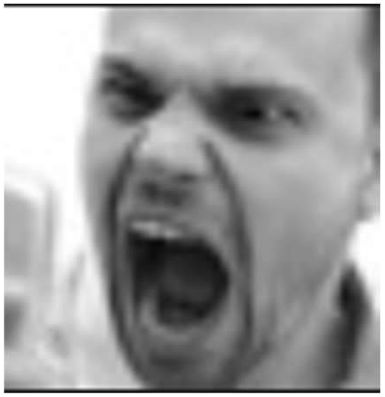
Goals:

- Apply PCA and Random Forest to the task of classifying facial emotions from static images
- Compare model performance with and without PCA to test whether dimensionality reduction improves classification
- Analyze which emotions are easier or harder to classify using accuracy rates and confusion matrices

DATASET OVERVIEW: FER-2013

- 48x48 grayscale facial images labeled across 7 emotions
- Folder structure already separated into training and testing sets
- Relevant because it directly matches our goal of emotion classification from static facial images
- Widely used benchmark dataset for facial expression recognition

FER-2013: 7 EMOTIONS



Angry

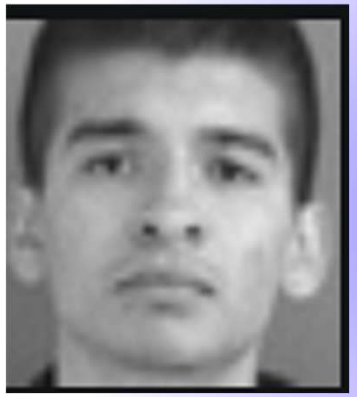
Disgust

Fear

Happy

Surprised

Sad



Neutral

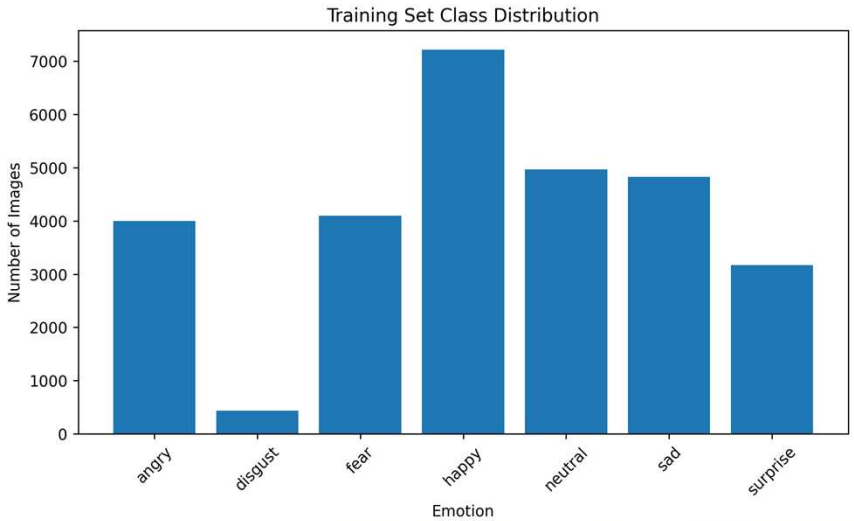
DATA EXPLORATION AND PREPROCESSING

Data Exploration:

- Class imbalance: happy has many more examples than disgust
- Visual Similarities
- Low Resolution: images make subtle facial features harder to detect

Preprocessing:

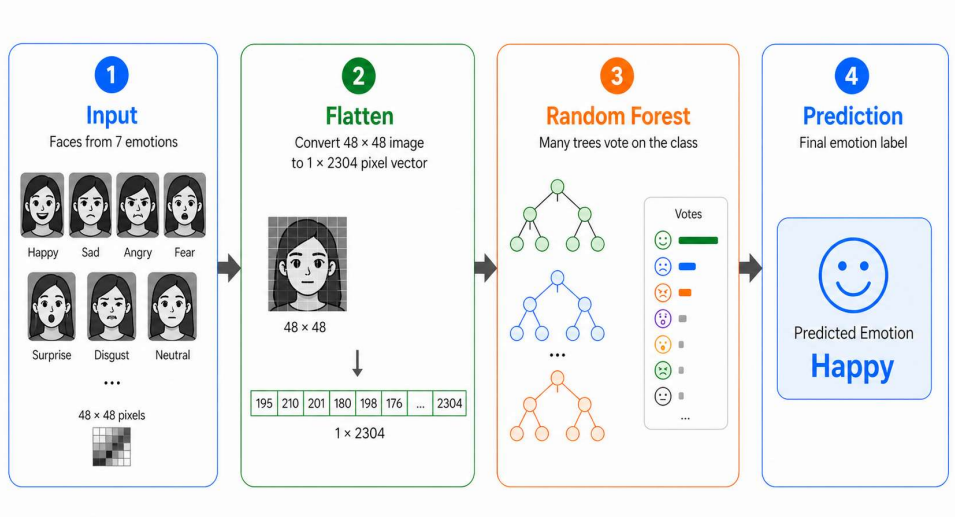
- Label encoding: mapped the seven emotions to integer labels from 0 to 6
- Normalization: scaled pixel values to the 0–1 range for consistency
- Flattening: converted each 48x48 image into a 2304-feature vector for Random Forest
- PCA: tested as an optional dimensionality reduction step before classification



METHODS COMPARED

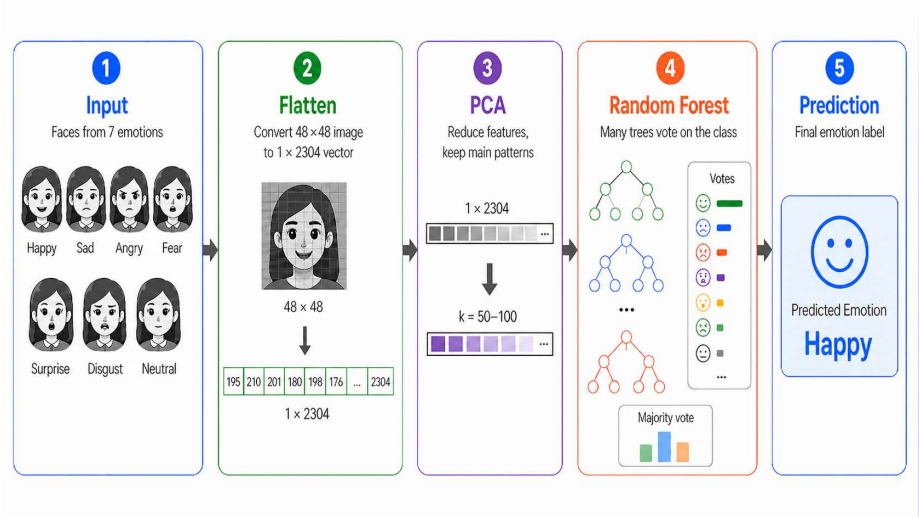
Method 1: Random Forest without PCA:

- Uses the full flattened image features
- Keeps subtle facial detail
- Larger/more dimensions and longer training time since it is using all features



Method 2: Random Forest + PCA:

- Reduces 2304 features into principal components (most important features)
- Can reduce noise and redundancy
- May remove useful emotion-related detail



Goal: Determine whether PCA improves emotion classification performance

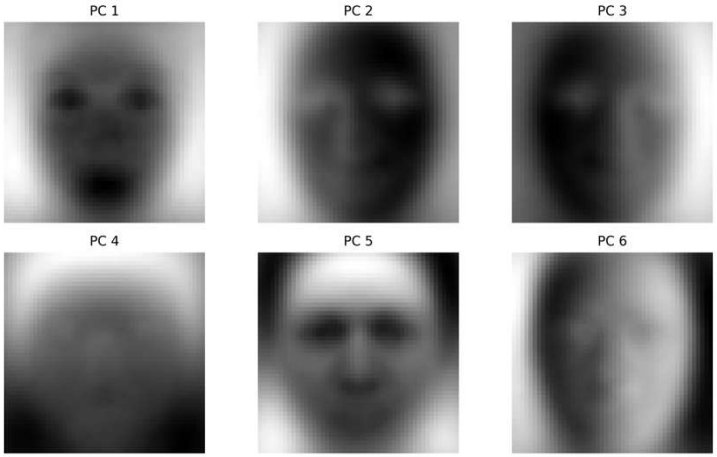
MODEL SETTINGS AND METRICS

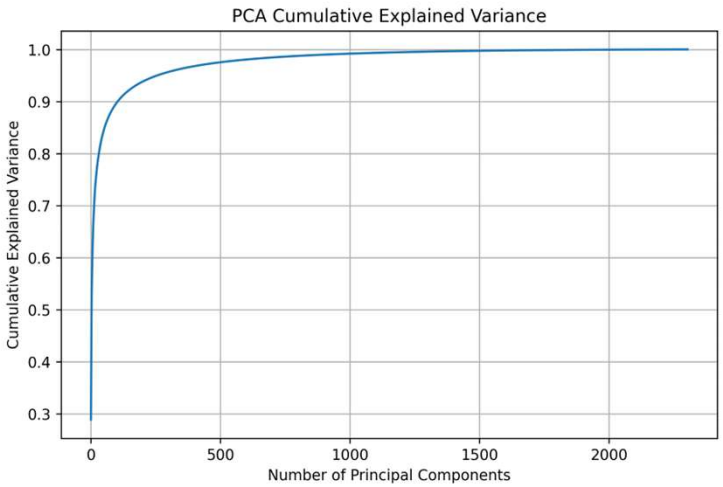
Parameter Settings:

- RandomForest: n_estimators = 100, random_state = 42, n_jobs = -1
- PCA: n_components = 50-300, random_state = 42

Classification metrics to evaluate the model:

- Accuracy: Proportion of all predictions correct
- Recall: How many correct out of total
- Precision: How correct was prediction
- F1: Balance between recall and precision
- Macro F1: for balanced performance across 7 emotions
- Confusion Matrix: to identify which emotions were often confused

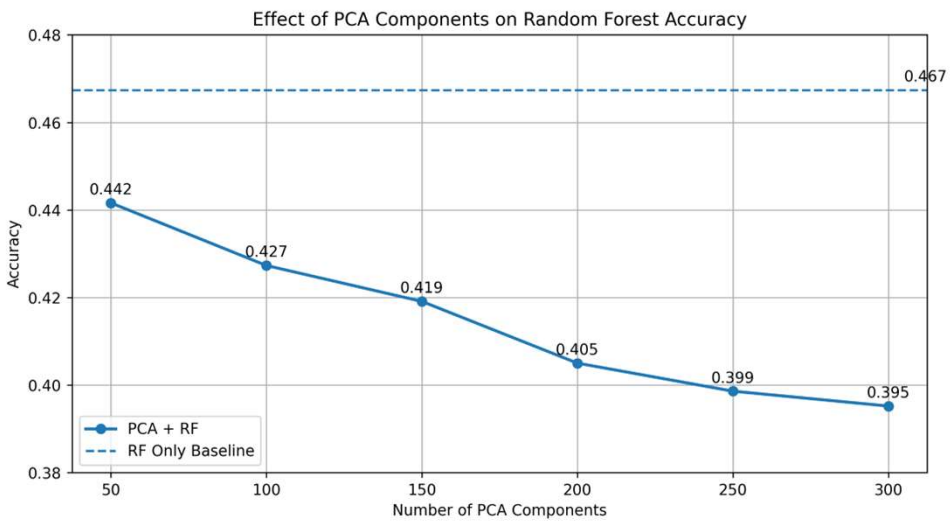


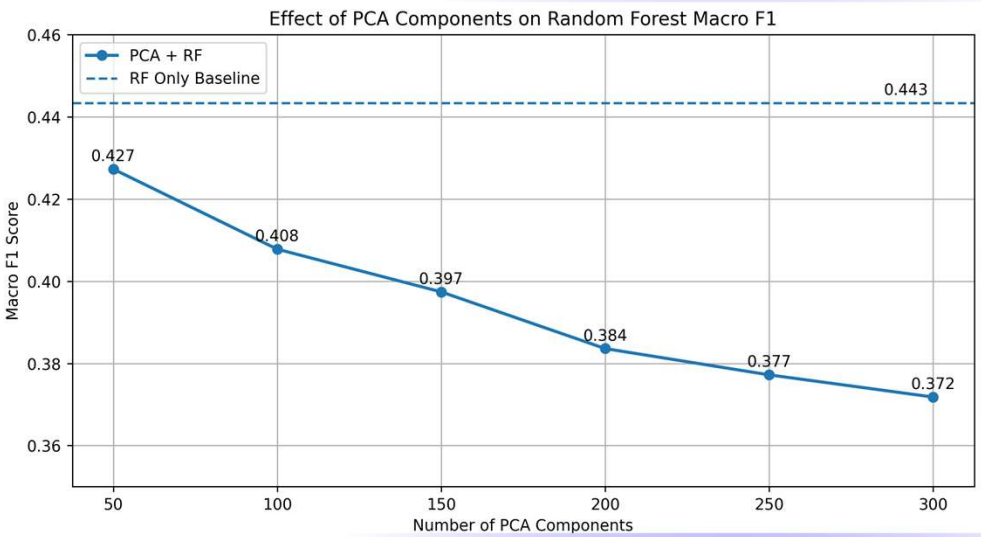


MODEL COMPARISON

Key Takeaways:

- Random Forest without PCA outperformed PCA at every component value
- As PCA components increased both metrics decreased
- PCA removed useful information rather than improving classification
- Variation patterns may not best separate emotion classes

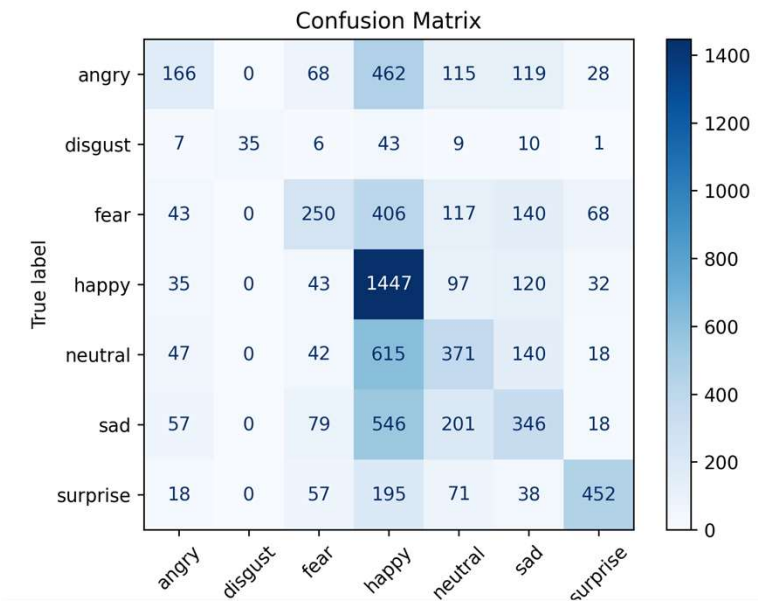


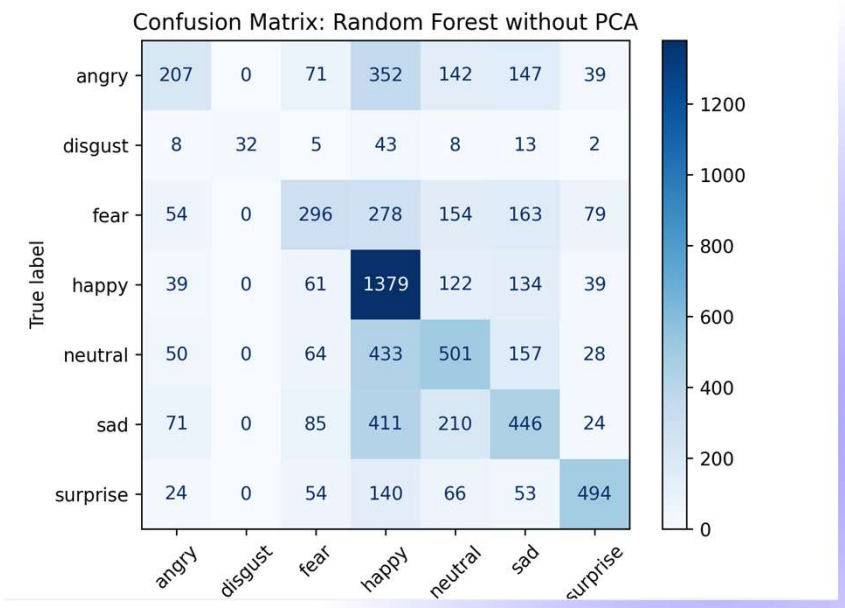


MODEL COMPARISON

Key Takeaways:

- Happy is strong in both models, but PCA has a slightly higher correct count for happy
- Neutral, sad, surprise, fear, and angry are generally better on the no-PCA model
- Both models struggle with disgust, likely because it has very little data
- Many classes are being predicted as happy, especially in the PCA model
- PCA seems to increase confusion toward the happy class for several emotions





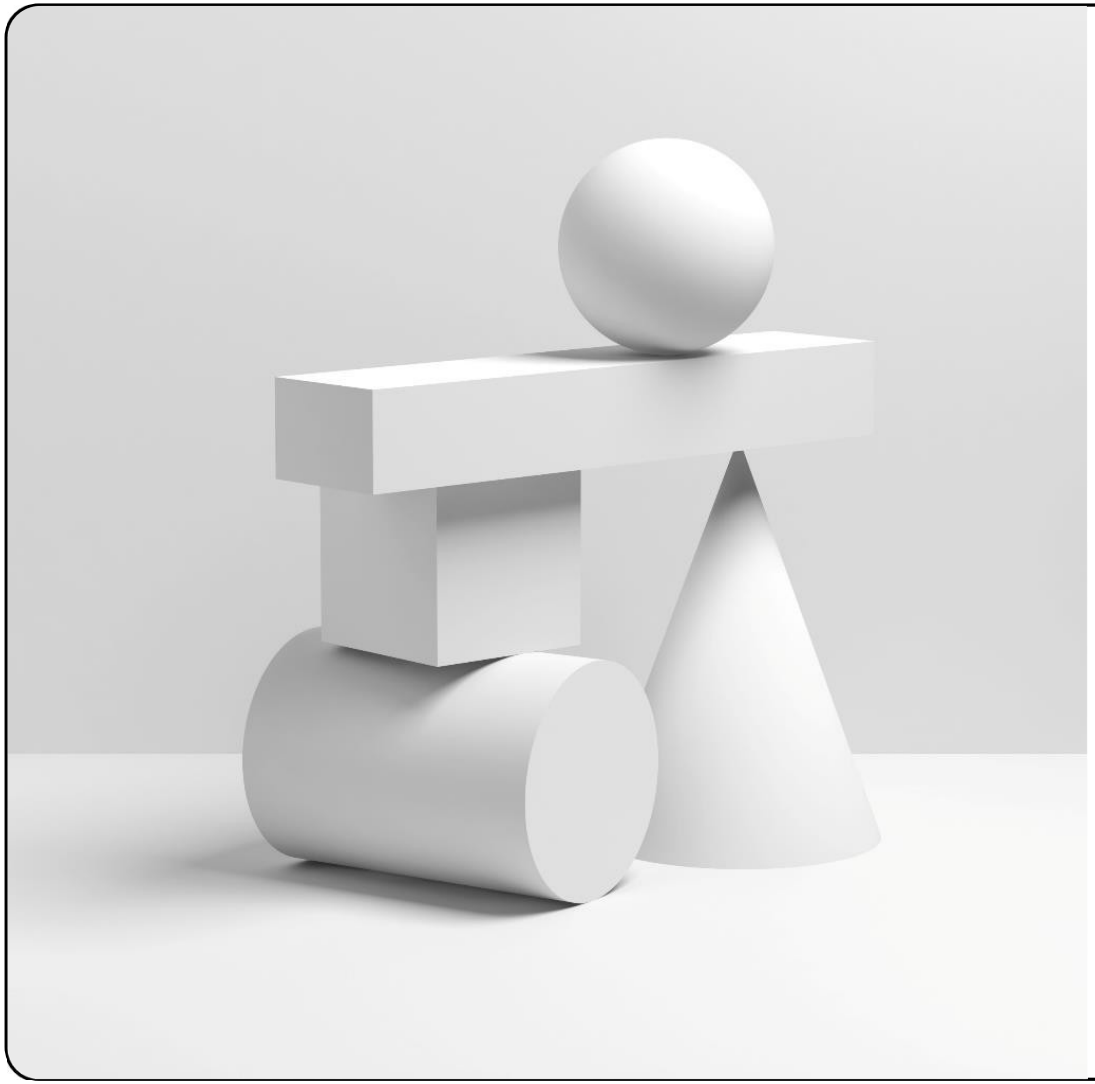
CONCLUSION AND IMPROVEMENTS

Key Takeaways:

- Random Forest without PCA performed better than Random Forest with PCA (simpler)
- Preserving full image detail was more useful than compressing the feature space
- PCA likely preserved overall variance, but removed subtle emotion-separating features
- Surprised was easier to identify because it had clearer visual cues and the most training examples
- Some emotions remained difficult because they were more visually subtle and overlapping(Happy and Angry)
- This shows that dimensionality reduction is not always beneficial, even when it makes the model simpler or faster

Future Work:

- Use class weighting or oversampling to improve minority classes such as disgust
- Compare against a CNN, which may better preserve the spatial structure of facial images
- Tune Random Forest hyperparameters such as max depth and class weight
- Grid Resolution



THANK YOU!